

# Parthajit Das

Kolkata, India

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## Education

**Indian Institute of Technology, Kharagpur, West Bengal, India**

July 2026 – Present

*M.Tech in Computer Science and Engineering*

**Techno Main Salt Lake, West Bengal, India**

Sept 2022 – June 2026

*B.Tech in Computer Science and Engineering (CGPA: 8.50)*

**Central Modern School, Baranagar, West Bengal**

*Indian School Certificate (ISC) – Class XII (Percentage: 90%)*

March 2022

*Indian Certificate of Secondary Education (ICSE) – Class X (Percentage: 94.2%)*

March 2020

## Experience

**Research Trainee**

June 2025 – July 2025

*Tenth Summer School on Computer Vision, Image Processing, Machine Learning Vision Language Models, Electronics and Communication Sciences Unit, Indian Statistical Institute (ISI), Kolkata*

- Selected for an intensive program covering machine learning, deep learning, and computer vision.
- Completed a mentored project on **Generative Adversarial Networks**.
- Awarded **Special Mention (Top 3)** among participating teams.

## Projects

**Hybrid Movie Recommendation System using Transformers and Dynamic Fusion**

*B.Tech Final Year Project*

- Designed a hybrid recommendation framework integrating **DistilBERT-based content embeddings**, **Neural Collaborative Filtering (NCF)**, **SVD**, and **LightGBM ranking**, supported by a dynamic fusion strategy to combine semantic and collaborative signals for personalized recommendations.
- Modeled complex user-item interactions using embedding layers and **MLP-based neural architectures**.
- Addressed key recommender system challenges including **cold-start**, **data sparsity**, and **filter bubble effects**.
- Utilized **MovieLens** and **TMDB datasets** for user ratings, movie metadata, and semantic feature extraction.
- Evaluated system performance using **RMSE**, **Precision**, **Recall**, and **F1-Score**, achieving superior results compared to individual recommendation models.

**Comparative Analysis of GAN Architectures**

*Tenth Summer School on Computer Vision, Image Processing, Machine Learning Vision Language Models*

- Implemented and compared **Vanilla GAN**, **WGAN**.
- Trained models on MNIST, CIFAR-10, and Anime datasets.
- Evaluated using **FID**, **Inception Score**, and **qualitative analysis**.
- Observed improved stability and performance in WGAN-GP over Vanilla GAN.

## Technical Skills

- **Programming:** C/C++, Python, Java, SQL
- **ML/DL:** PyTorch, TensorFlow, Transformers, LightGBM
- **Libraries:** NumPy, Pandas, Matplotlib
- **Tools:** Git, GitHub, Kaggle, Google Colab, MySQL

## Achievements

- **Special Mention Award (Top 3)** – ISI Kolkata Summer School  
Recognized among the top projects for “*Comparative Analysis of Generative Adversarial Network Architectures*”.
- GATE CSE 2025 and 2026 Qualified
- 100+ coding problems in leetcode across LeetCode and GeeksforGeeks

## Publications / Manuscripts

• **An Ensemble Learning Framework for Hybrid Movie Recommendation Systems Using Transformers**

Developed a research-oriented hybrid recommender system combining DistilBERT, Neural Collaborative Filtering (NCF), SVD, and LightGBM to improve recommendation accuracy and mitigate cold-start and data sparsity issues. Conducted comparative evaluation using RMSE, Precision, Recall, and F1-Score, demonstrating superior performance over individual recommendation models.

Link: [View Manuscript](#)

• **Comparative Analysis of Generative Adversarial Network Architectures**

Manuscript prepared during research training at ISI Kolkata Tenth Summer School, involving implementation and evaluation of GAN variants (Vanilla GAN, WGAN, WGAN-GP).

Link: [View Manuscript](#)